

STAVANGER VAALAND, Norway

WMO: 014160

Lat: 58.9574N

Lon: 5.7287E

Elev: 72

StdP: 100.46

Time Zone: 1.00 (EUC)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-6.1	-4.7	-14.1	1.1	-3.1	-12.0	1.3	-2.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.6	24.6	16.8	22.3	15.8	20.4	15.0	17.8	22.7	16.8	20.7	15.9	19.2	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.0	11.5	19.0	15.1	10.8	18.2	14.3	10.3	17.4	50.4	22.8	47.5	20.8	44.8	19.1	21.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
N/A	N/A	N/A	DB	-6.6	28.4	2.5	2.8	-8.4	30.5	-9.9	32.1	-11.3	33.7	-13.1	35.8	
			WB	-7.7	19.3	2.7	1.2	-9.6	20.2	-11.2	20.9	-12.7	21.6	-14.6	22.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.5	1.9	2.0	4.1	7.5	10.6	13.1	15.7	15.5	13.3	9.1	5.8	3.4
	DBStd	5.70	3.07	3.12	2.59	2.95	3.70	2.52	2.88	2.15	2.40	2.51	3.10	3.43
	HDD10.0	1161	250	224	184	86	34	2	0	0	3	44	129	204
	HDD18.3	3598	508	457	443	323	241	159	93	94	154	286	377	463
	CDD10.0	629	0	0	0	12	54	96	177	169	101	18	2	0
	CDD18.3	24	0	0	0	0	2	3	13	5	1	0	0	0
	CDH23.3	123	0	0	0	0	14	14	74	16	4	0	0	0
	CDH26.7	19	0	0	0	0	2	1	16	1	0	0	0	0
Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Precipitation	PrecAvg	1199	103	75	72	56	65	69	90	122	143	144	139	120
	PrecMax	1733	236	177	200	126	168	161	244	241	328	299	244	248
	PrecMin	841	4	1	2	3	2	16	21	5	27	37	25	8
	PrecStd	193	60	43	46	26	35	33	52	52	56	61	56	60
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.4	9.2	12.0	19.5	25.2	25.5	28.5	25.6	22.7	17.8	13.7	10.2
		MCWB	6.8	6.5	8.1	11.0	16.1	16.1	18.7	17.0	16.5	13.8	10.7	9.0
	2%	DB	7.6	7.6	9.6	16.5	22.1	21.5	25.2	22.2	19.7	15.2	11.7	9.4
		MCWB	6.6	6.0	6.6	9.7	14.1	15.3	17.3	15.9	14.3	12.2	9.9	8.3
	5%	DB	6.9	6.9	8.1	14.3	19.2	19.1	22.8	20.5	18.2	13.6	10.5	8.6
		MCWB	6.0	5.6	5.9	8.8	12.9	13.8	16.6	15.3	13.8	11.0	8.8	7.4
	10%	DB	6.2	6.2	7.3	12.3	16.7	17.3	20.5	19.1	16.8	12.5	9.8	7.8
		MCWB	5.1	4.9	5.3	7.9	11.5	13.0	15.8	14.7	13.3	10.2	8.3	6.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.7	7.2	9.0	11.5	16.8	17.2	19.6	18.2	17.4	14.4	11.7	9.5
		MCDB	8.0	8.0	10.8	17.1	23.9	22.2	26.3	22.3	22.0	16.3	12.3	9.9
	2%	WB	7.0	6.5	7.4	10.2	15.0	15.7	18.4	17.0	15.7	13.0	10.4	8.6
		MCDB	7.3	7.3	8.9	15.1	20.9	20.6	24.1	20.5	18.0	14.7	11.4	9.2
	5%	WB	6.2	5.9	6.5	9.4	13.4	14.6	17.4	16.1	14.6	11.8	9.5	7.7
		MCDB	6.6	6.7	7.8	13.4	18.2	18.4	21.8	19.0	17.4	13.1	10.5	8.4
	10%	WB	5.1	5.1	5.8	8.5	12.1	13.6	16.4	15.4	13.7	10.8	8.4	6.6
		MCDB	6.0	6.0	6.9	11.6	16.1	16.6	19.9	18.2	16.1	12.0	9.5	7.4
Mean Daily Temperature Range	5% DB	MDBR	3.4	4.1	4.9	6.7	7.1	6.9	6.6	6.1	5.5	4.7	3.7	3.4
		MCDBR	3.9	4.2	5.7	9.5	11.0	9.7	10.2	8.0	7.4	5.9	4.0	3.6
		MCWBR	3.9	3.5	3.7	4.8	5.3	4.6	4.3	3.8	3.8	3.9	3.5	3.8
	5% WB	MCDBR	3.9	4.0	5.2	8.5	10.5	8.7	9.2	6.8	6.6	4.8	4.0	3.9
		MCWBR	4.2	3.7	3.8	4.6	5.3	4.7	4.2	3.9	4.0	4.2	4.1	4.2
Clear-Sky Solar Irradiance	taub	0.299	0.324	0.344	0.376	0.366	0.360	0.374	0.374	0.355	0.335	0.305	0.277	
	taud	2.177	2.282	2.321	2.299	2.418	2.464	2.443	2.441	2.483	2.428	2.263	2.120	
	Ebn at Noon	496	689	792	826	866	876	853	824	782	678	494	383	
	Edh at Noon	47	70	90	109	104	101	101	94	78	61	43	35	
All-Sky Solar Radiation	RadAvg	0.30	0.90	2.13	3.82	5.12	5.53	4.90	3.78	2.43	1.17	0.41	0.19	
	RadStd	0.05	0.16	0.26	0.40	0.59	0.62	0.50	0.32	0.22	0.15	0.06	0.02	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (30 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-116	N/A	N/A

Nomenclature: See separate page